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San Leandro, CA 94578

June 9th, 2025

GRAS Notification Program
Office of Food Additive Safety, HFS-200
5001 Campus Drive
College Park, MD 20740-3835

GRAS Notice for *Saccharomyces cerevisiae* strain BY-927

Dear Sir or Madam,

We respectfully submit the attached GRAS Notice for Berkeley Fermentation Science Inc.'s *Saccharomyces cerevisiae* strain BY-927. BY-927 is intended for use in brewing to improve beer quality by preventing the formation of the off-flavor compound diacetyl. More detailed information regarding product identification, intended use levels, the manufacturing process, and safety of the strain is set forth in the attached GRAS Notice.

Berkeley Fermentation Science Inc. has determined that BY-927 is GRAS for its intended use based on scientific procedures in accordance with 21 C.F.R. §170.30. Therefore, the use of BY-927, as described in this GRAS Notice, is exempt from the requirement of premarket approval as set forth in the Federal Food, Drug, and Cosmetic Act. Information supporting the GRAS status of BY-927, under the intended conditions of use, is enclosed for review by the Agency.

Should you have any questions or concerns regarding this GRAS Notice, please do not hesitate to contact me at any point during the review process by email at charles@berkeleyyeast.com. Thank you for your time and your consideration.

Sincerely,

Berkeley Fermentation Science Inc.

By: _____
Name: Charles Denby, Ph.D.
Title: Chief Executive Officer

GRAS Notice for *Saccharomyces cerevisiae* strain BY-927

PREPARED FOR:

Office of Food Additive Safety (FHS-200)
Center for Food Safety and Applied Nutrition
Food and Drug Administration
5001 Campus Drive
College Park, MD 20740-3835

DATE:

June 9, 2025

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PART 1: SIGNED STATEMENTS AND CERTIFICATION

1.1. Exemption from premarket approval

In accordance with Title 21 C.F.R. §170.30, Berkeley Fermentation Science Inc. hereby informs the United States (U.S.) Food and Drug Administration (FDA) of the view that its *Saccharomyces cerevisiae* strain BY-927 is Generally Recognized as Safe (GRAS) under the conditions of its intended use and is, therefore, exempt from the requirement for premarket approval from the Federal Food, Drug, and Cosmetic Act.

1.2. Basis for GRAS determination

The determination of the GRAS status is based on scientific procedures and conforms to the regulations in accordance with Title 21 C.F.R. §170.30.

1.3. Name and address of the notifier

Berkeley Fermentation Science Inc.
15555 E 14th Street, Suite 525
San Leandro, CA 94578

1.4. Name of the notified substance

The notified substance is a *Saccharomyces cerevisiae* strain called BY-927. BY-927 expresses a gene encoding an α -acetolactate decarboxylase derived from *Acetobacter aceti*.

1.5. Intended conditions of use

Yeast strain BY-927 is intended for use in brewing to improve beer quality by preventing the formation of the vicinal diketone diacetyl. Diacetyl is a small molecule produced as a byproduct of yeast metabolism during beer fermentation. If present at concentrations above the threshold of human detection, diacetyl will impart an undesirable buttery flavor. BY-927 expresses α -acetolactate decarboxylase (ALDC), an enzyme that converts α -acetolactate, the chemical precursor of diacetyl, into acetoin. Expression of ALDC prevents the accumulation of α -acetolactate and the formation of diacetyl, and keeps diacetyl concentrations below sensory threshold levels.

1.6. Availability of information for FDA review

Berkeley Fermentation Science Inc. undertakes that all data and information used in this GRAS determination for use of BY-927 yeast in brewing will be made available for the FDA to review and copy at the following address:

Berkeley Fermentation Science Inc.

Attention: Charles Denby
15555 E 14th Street, Suite 525
San Leandro, CA 94578
USA

Data will be sent to the FDA upon request.

No information used in this part of this notification is trade secret or confidential commercial information.

1.7. Disclosure and certification

No data or information contained in parts 2 through 7 of this GRAS notice are exempt from disclosure under the Freedom of Information Act, 5 U.S.C. 552.

To the best of our knowledge, this GRAS notice for *Saccharomyces cerevisiae* strain BY-927 and its use in brewing is a complete, representative, and balanced submission that includes both favorable and unfavorable information, known to Berkeley Fermentation Science Inc. and pertinent to the evaluation of the safety and GRAS status of the use of the substance.

Charles Denby, Ph.D.	Date
Chief Executive Officer	
Berkeley Fermentation Science Inc.	

PART 2: IDENTITY, METHOD OF MANUFACTURE, SPECIFICATION, AND PHYSICAL OR TECHNICAL EFFECT

2.1. Identity of the notified substance

The subject of this notification is an industrial brewing *Saccharomyces cerevisiae* strain called BY-927. BY-927 is intended for use in brewing to improve beer quality by keeping diacetyl, an off-flavor compound produced by yeast with a buttery, butterscotch-like flavor, below sensory threshold levels. This strain contains a gene from *Acetobacter aceti* that encodes an α -acetolactate decarboxylase enzyme (ALDC) that converts the diacetyl precursor, α -acetolactate, directly into acetoin.

2.2. Intended use of BY-927

Berkeley Fermentation Science Inc. has proposed the use of BY-927 in brewing to prevent the formation of the vicinal diketone diacetyl (IUPAC name butane-2,3-dione). Diacetyl is a small molecule produced as a byproduct of yeast metabolism during beer fermentation¹⁻³. If present at concentrations above the threshold of human detection, diacetyl will impart an undesirable buttery flavor. The biochemical pathway that leads to diacetyl production during beer fermentation is outlined in Figure 1.

The chemical precursor of diacetyl is α -acetolactate, which is produced by actively fermenting yeasts as a metabolic intermediate in the valine biosynthesis pathway. A proportion of the α -acetolactate produced is excreted from the cell, where it is spontaneously decarboxylated to produce diacetyl⁴. Due to the relatively low flavor detection threshold for diacetyl, even small quantities that are produced by this process can negatively impact the flavor and aroma of beer⁵.

The detrimental impact of diacetyl on beer flavor has motivated the brewing community to develop strategies to minimize its concentration in beer prior to packaging⁵. Two predominant strategies exist for this. The first is to perform a 'diacetyl rest', in which brewers leave the yeast and beer in the fermenter for 1-30 days after fermentation. During this period of days or weeks, the yeast will slowly uptake diacetyl from the beer and reduce it first to acetoin, then again to butane-2,3-diol. A diacetyl rest leads to a gradual elimination of diacetyl from the beer, and the diacetyl concentration will eventually fall below the flavor detection threshold, at which point the beer can be packaged. Butane-2,3-diol has a relatively high flavor detection threshold and does not impact the final aroma or taste of the beer. As the yeast are no longer vigorously fermenting during this period, very little additional diacetyl is produced. However, the yeast may not be active enough to metabolize all diacetyl present, and any remaining α -acetolactate still has the potential to become diacetyl after the beer is packaged. In addition, the added time increases the cost of brewing.

A second commonly employed method that brewers use to manage diacetyl is to add an α -acetolactate decarboxylase (ALDC) enzyme to their beer fermentations^{6,7}. ALDC directly converts α -acetolactate to acetoin⁸, thereby circumventing diacetyl formation entirely (Figure 1). Several purified ALDC enzymes derived from food-grade bacteria are commercially available for this purpose^{9,10}. However, due to its unpredictable performance, most breweries do not supplement their fermentations with ALDC, or do so in concert with a diacetyl rest. Additionally, if the enzyme degrades before α -acetolactate has been secreted out of the cell, then diacetyl may still form in packaged beer.

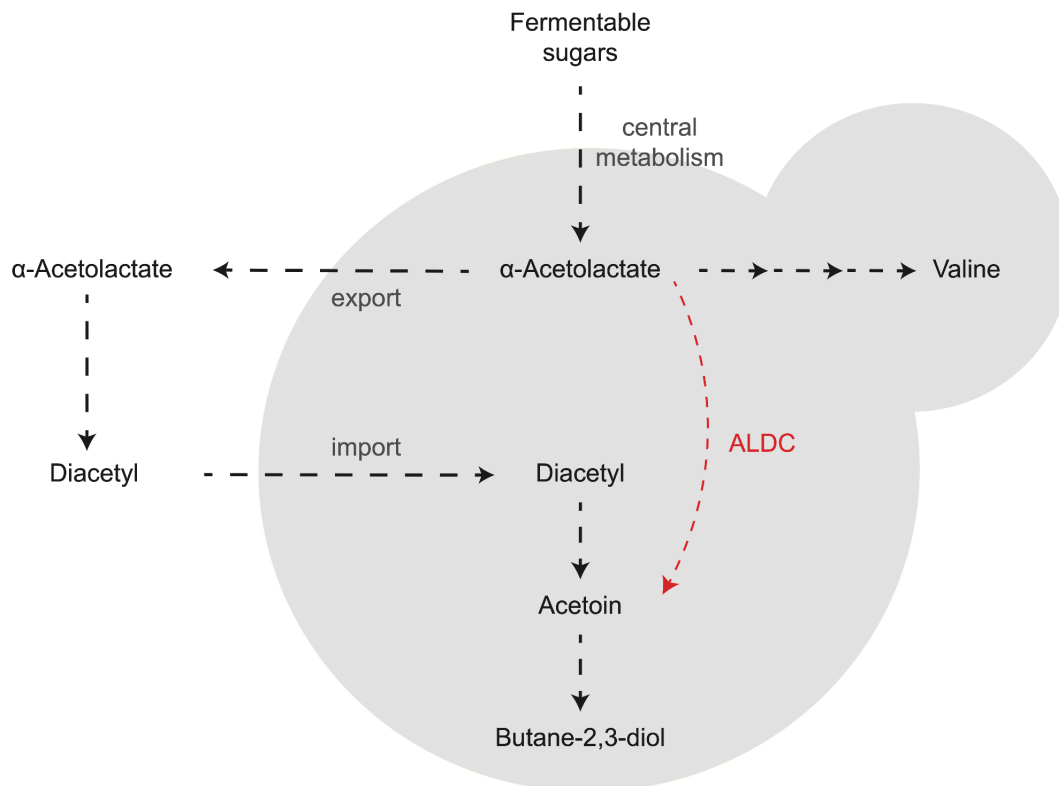


Figure 1. Biochemical pathway of diacetyl formation and degradation during brewing. Diacetyl is formed naturally during the brewing process. Fermentable sugars are consumed by yeast through glycolysis, and pyruvate is converted to α -acetolactate, a metabolic intermediate in the valine biosynthesis pathway. Excess α -acetolactate accumulates and slowly forms diacetyl through non-catalyzed decarboxylation. During a diacetyl rest, diacetyl is reduced to acetoin and then to butane-2,3-diol, both flavor-benign molecules. ALDC converts α -acetolactate directly into acetoin.

The practice of dry hopping - adding flavoring hops directly to beer in the fermentation tank - also has a significant impact on the production of diacetyl in beer. The hop cones used for dry hopping contain amylase enzymes that hydrolyze oligosaccharides to release shorter, more readily fermentable sugars. When added to beer fermentations, these amylases hydrolyze non-fermentable oligosaccharides in the wort and release shorter chain sugars that yeast can readily ferment. This release of short-chain sugars re-initiates yeast fermentation, leading to the production of additional diacetyl. Dry hopping thus creates an additional challenge for brewers

concerned with diacetyl production, and it can further extend brewing timelines by forcing an even longer diacetyl rest^{11,12}. Addition of purified ALDC at the same time as dry hopping can help minimize this hop-catalyzed diacetyl release, but as mentioned above, the efficacy of ALDC addition is highly variable across different fermentations.

Because neither method is completely effective, significant efforts have been made to genetically engineer brewers yeast strains that either produce less diacetyl, or remove diacetyl by converting it to butane-2,3-diol. A variety of genetic strategies towards these ends have been explored. These include: 1) lowering the activity of genes that produce alpha-acetolactate^{13–16}, 2) increasing the activity of genes that convert alpha-acetolactate to subsequent metabolites in the valine biosynthesis pathway¹³, 3) increasing the activity of genes that convert diacetyl to acetoin and butane-2,3-diol¹³, and 4) heterologous expression of ALDC in yeast^{17,18}. Of these, ALDC expression in yeast has most consistently been shown to reduce the concentration of diacetyl in beer^{17,19–26}.

To create an ALDC-expressing brewer's yeast strain that would shorten the brewing timeline and obviate the need for purified ALDC, we engineered BY-927 to express the ALDC from *Acetobacter aceti*. Production of the ALDC enzyme inside the cell is a reliable method that is robust to process variation, resulting in a yeast strain that does not accumulate α -acetolactate and finished beer with diacetyl below sensory threshold levels.

BY-927 is used as any other commercial yeast would be used in the beer manufacturing process.

2.3. Host microorganism

2.3.1. History of use

Saccharomyces cerevisiae has been used in food and beverage production for thousands of years. The first evidence of beer production comes from ancient Iran, Iraq, and Egypt, although spontaneous fermentations resulting in beer were likely happening far earlier than that, as wild yeasts in the air will ferment most grain left unattended²⁷. Similarly, archaeological evidence of wine production dates to 5400 BC Iran²⁸, but spontaneous fermentations of grapes into wine by wild yeasts were likely happening even earlier. Ancient peoples utilized yeast in the production of other fermented foods and baked goods²⁹. Today yeast is an integral component in many industries, and millions of tons are produced commercially each year. The food and beverage applications of *S. cerevisiae* range from alcoholic beverages to bread, coffee, and chocolate, to name a few. *S. cerevisiae* has also served as a model organism in scientific research and was the first eukaryote to have its genome completely sequenced.

The parent yeast strain from which BY-927 was derived is an industrial brewing strain of *Saccharomyces cerevisiae* that has been used extensively in commercial beer production for decades. Like many industrial brewing strains, the parent yeast strain is tetraploid.

2.3.2. Taxonomy

The parent strain is classified as belonging to the *Saccharomyces cerevisiae* species. A considerable amount of genetic diversity exists within the species *S. cerevisiae*, and genome sequencing efforts have covered thousands of individuals³⁰. The strains used for various industrial food and beverage manufacturing processes have evolved considerably from their wild ancestors, which likely inhabited primeval forests³¹. These industrial strains have been categorized into distinct clades: one containing sake yeasts, one containing wine yeasts, a mixed clade containing bread yeasts, and two separate beer yeast clades³⁰.

2.4. Donor organisms

2.4.1. Taxonomy

The following organisms contributed DNA to BY-927:

Acetobacter aceti

BY-927 contains a gene that encodes an ALDC derived from *Acetobacter aceti*, a gram-negative bacterium belonging to the acetic acid bacteria family and a food microorganism used in commercial vinegar production³². *Acetobacter* species oxidize ethanol to form acetic acid, the main chemical component of vinegar. *A. aceti* has a long history of safe use in the production of vinegar, which dates back more than 5,000 years.

Saccharomyces cerevisiae

BY-927 contains promoter and terminator DNA sequences derived from the *S. cerevisiae* strain S288C. The history of use of this species is given in Section 2.2.1. The S288C strain, a widely used lab strain, was isolated through genetic crosses for biochemical studies by Robert Mortimer³³ and serves as the reference genome sequence in the *Saccharomyces* Genome Database. S288C is mainly derived from a wild yeast strain that was isolated from figs; minor portions of its genetic material are derived from baking yeasts and other yeasts isolated from fruit. In addition, the sequences isolated from S288C are 99% identical to the sequences found in the parent strain.

2.4.2. Genetic material from donor organisms

The ALDC gene sequence

The ALDC gene sequence encodes a decarboxylase that converts α -acetolactate to acetoin^{34–36}. Yeast biosynthesizes α -acetolactate as an intermediate in the valine biosynthesis pathway. During fermentation, excess α -acetolactate is secreted into the wort, where it is spontaneously decarboxylated to produce diacetyl⁴, which imparts an undesirable butter-like flavor to finished beer. ALDC converts α -acetolactate directly into acetoin, thereby preventing diacetyl formation

and ensuring that excess α -acetolactate does not accumulate and form diacetyl in packaged beer.

The *A. acetii* ALDC gene was previously shown to reduce diacetyl production when expressed in brewer's yeast^{23–25}. BY-927 expresses a truncated variant of the originally identified *A. acetii* ALDC gene, as this variant has been found to possess greater activity than the original sequence²³.

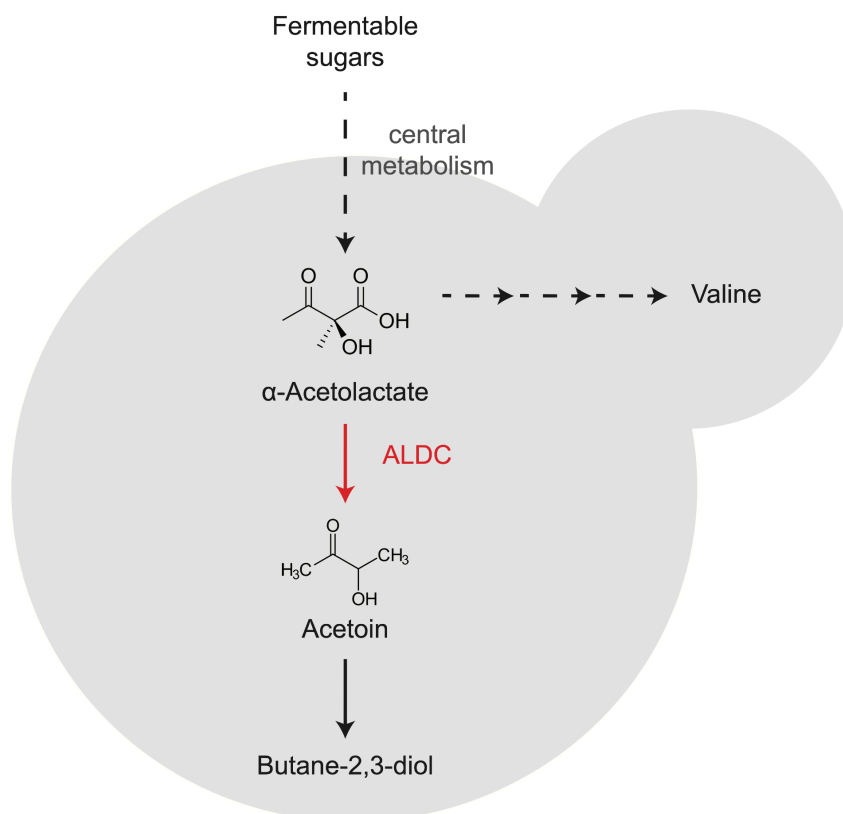


Figure 2. Introduction of ALDC prevents diacetyl formation. ALDC converts α -acetolactate into acetoin, preventing diacetyl formation. A yeast-endogenous dehydrogenase enzyme encoded by BDH1 reduces acetoin to butane-2,3-diol.

The promoter sequence

The promoter sequence (i.e. the genetic element immediately 5' of the integrated gene) is derived from *S. cerevisiae* strain S288C. Promoter sequences define where transcription of a gene by RNA polymerase begins. The promoter used in this strain consists of a 800 bp fragment ending at the base pair immediately preceding the ATG start codon of its associated gene.

The terminator sequence

The terminator sequence (i.e. the genetic element immediately 3' of the integrated gene) is derived from *S. cerevisiae* strain S288C. Terminator sequences provide a signal that triggers the release of mRNA from the transcription complex during gene expression. The terminator used in this strain consists of a 225 bp fragment starting immediately after the stop codon of its associated gene.

2.5. The modified microorganism

2.5.1. General construction strategy

The strain was constructed with a single genomic integration. The molecular tools and practices used during the construction of BY-927 are standard to the field of molecular biology and yeast genetics.

2.5.1.1. Integration of the ALDC insert

First, we generated the construct containing the genetic material described in Section 2.4.2 using standard methods of genetic cloning as follows: the gene coding for the ALDC enzyme was synthesized by oligonucleotide assembly and inserted into a cloning vector, which was then sequenced. The promoter and terminator sequences were amplified by PCR from *S. cerevisiae* strain S288C genomic DNA and inserted into cloning vectors, which were then sequenced.

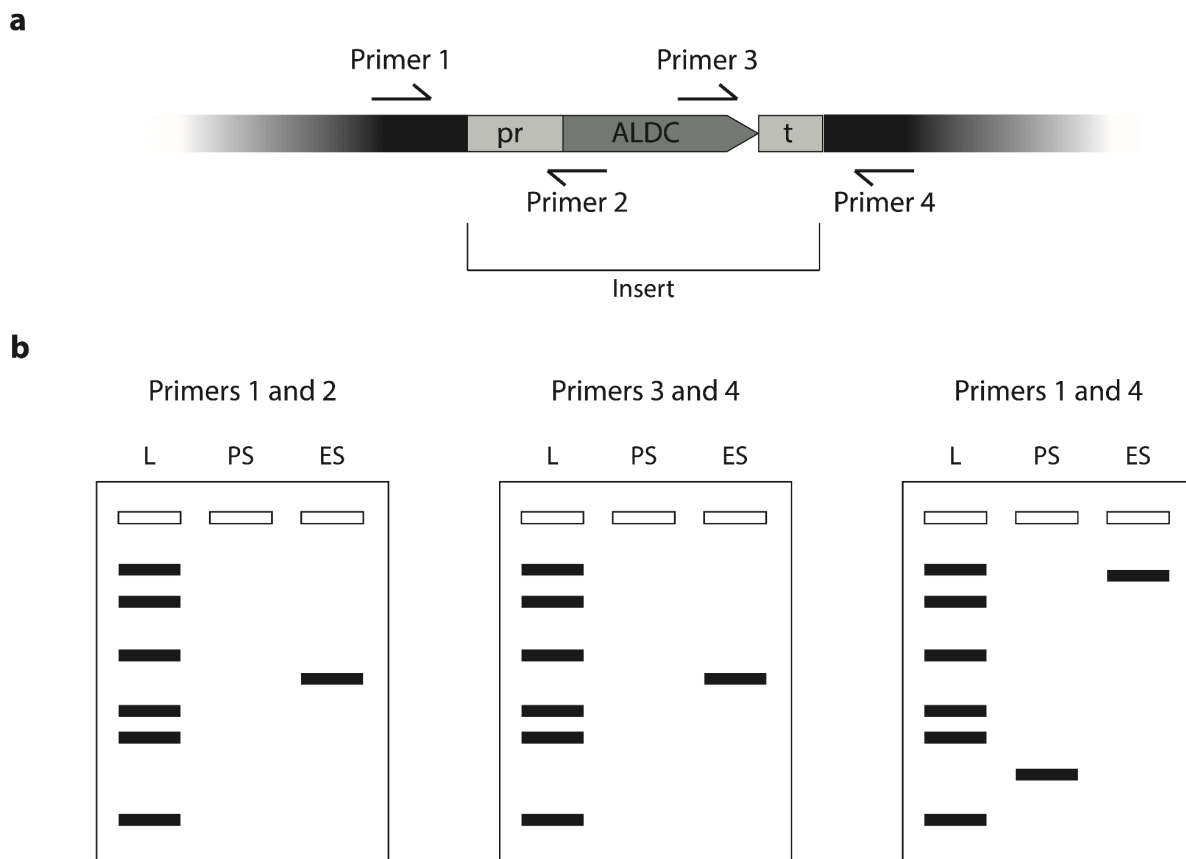
After each plasmid was sequenced, further cloning to generate the insert was performed by Golden Gate Assembly, a restriction enzyme-based cloning method that avoids mutations introduced by PCR-based methods³⁷. The promoter, gene, and terminator plasmids were assembled into a single plasmid using Golden Gate Assembly. The resulting plasmid was sequenced to confirm that it contained all three genetic elements and then linearized using a restriction enzyme.

The resulting DNA fragment containing the ALDC gene under the regulation of a native *S. cerevisiae* promoter and terminator, was directly integrated into the genome of the parent strain at a single locus.

In addition to the linear fragment, transformation of the parent strain also used an integration aid plasmid. This plasmid encodes the tools for genetic incision, including an endonuclease that specifically cuts the target locus. The plasmid also encodes an antibiotic resistance gene that confers resistance to G418, which allows the cell to maintain the plasmid through selection by growth on G418-containing medium. This plasmid was used only as a co-transformation aid and none of its genetic material was integrated into the cell. Following confirmation of the integration, the plasmid was removed by passaging on YPD medium.

2.5.2. Screening method for transformants

The transformed strain was plated onto G418-containing medium in order to select transformed cells. Confirmation of the integration event at the single locus across four homologous chromosomes was conducted by diagnostic PCR (Figure 3).



2.5.3. Genetic characterization of the modified microorganism

2.5.3.1. The loss of the plasmid containing the antibiotic resistance gene

Following confirmation of the integration event, the integration aid plasmid was removed from the engineered strain by passaging on YPD medium. Loss of the plasmid was confirmed by testing for growth on YPD medium containing the antibiotic G418 and confirming sensitivity. BY-927 is unable to grow on a G418-containing medium, like the non-engineered parent strain. Therefore, any antibiotic resistance genes were removed from the final strain and no antibiotic resistance is conferred to BY-927.

2.5.3.2. Genetic stability of the modifications

The inserted DNA is integrated into the *S. cerevisiae* genome at the target locus across four homologous chromosomes. In previous research, we have demonstrated that homozygosity is critically important for integration stability³⁸. If engineering results in heterozygosity, the modifications are highly unstable, often reverting to the parental allele within only a few generations. On the other hand, integration across four homologous chromosomes is stable over hundreds of generations.

We confirmed the genetic stability of the strain over 100 generations by serially passaging BY-927 on YPD medium. After passaging, the presence of the inserted DNA was re-confirmed by diagnostic PCR using the parent strain as a control.

2.5.3.3. The introduced DNA is the expected sequence

To confirm that the correct sequence was inserted, we used PCR to amplify the locus containing the ALDC insert with primers that bind outside the engineered region (represented by primers 1 and 4 in Figure 3b). The resulting DNA amplicon was sequenced and compared to an expected reference sequence of the engineered region. The actual DNA sequence and the intended reference sequence matched exactly, thus confirming that the engineering work produced the expected outcome.

2.5.4. BY-927 and its parent strain are highly genetically similar

The genome of all *S. cerevisiae* strains contain long terminal repeat sequences known as δ elements. These δ elements are the remnants of Ty1 and Ty2 transposon integration events³⁹. The number and location of these δ elements are strain-specific and serve as an effective means of distinguishing between strains of *S. cerevisiae*⁴⁰. We verified the genetic relationship between the parent strain and the engineered strain by amplifying these δ sequences by PCR with primers 5'-GTGGATTTTATTCCAAC-3' and 5'-TCAACAATGGAATCCCAAC-3' (delta2/delta12 primer pair), a technique called interdelta analysis. PCR amplification resulted in identical banding patterns from both strains. We therefore conclude that the engineered strain and its parent strain are highly genetically similar.

2.6. Method of manufacture of BY-927

2.6.1. Overview of manufacturing method

BY-927 is manufactured in accordance with Good Manufacturing Practices. Production is conducted at a fermentation facility with established procedures and equipment suitable for large-scale contained production of *S. cerevisiae*. At all steps, appropriate analyses are conducted to ensure that the yeast meets the finished product specifications.

BY-927 is started from frozen glycerol stocks by streaking directly onto YPD agar plates, grown in liquid medium in increasing volumes, and inoculated into a propagation vessel with aeration. Quality control specifications are measured throughout the process; if a lot does not meet target specifications, then the batch is rejected. When the culture has depleted all fermentable sugar, the yeast cell mass is collected as a concentrated liquid slurry.

2.6.2. Raw materials

All raw materials and processing aids used in the manufacture of BY-927 are used in accordance with U.S. regulations, are the subject of an effective food contact notification, or are concluded to be GRAS for the intended use. The raw materials are standard food-grade ingredients used in traditional yeast production and include malt extract or molasses (the carbon source) and other essential nutrients. The only other material used is a food-safe, 100% vegetable oil-based defoamer, which is commonly used in commercial beer production. None of the components used in the manufacturing process or in the final ingredient include or are derived from one of the nine major food allergens (milk, eggs, fish, Crustacean shellfish, tree nuts, peanuts, wheat, soybeans, or sesame).

2.6.3. Lab stage

The manufacturing process begins with a series of carefully controlled serial steps with cultures being used to inoculate successively larger volumes. Fermentation is performed under controlled conditions. A brief overview of these steps is listed below.

1. Master stock cultures of yeast strains are stored in cryogenic conditions at -80°C in glycerol.
2. Propagation begins by inoculating BY-927 onto a YPD (Yeast extract, Peptone, Dextrose) agar plate from frozen master stocks with a single-use sterile loop.
3. After growth, colonies are transferred into 40 mL YPD liquid medium and grown for 24 hours.
4. The 40 mL pre-culture is sequentially transferred into increasing volumes in propagation medium (1L, 20 L; see Section 2.6.4 for a description of the ingredients).
5. Microbial QC is performed on the final starter culture.

2.6.4. Propagation and recovery

The yeast from the lab is inoculated into propagation vessels, grown until it reaches peak cell count, and harvested. A brief overview of these steps is listed below.

1. To prepare the propagation medium, malt extract or molasses is mixed with water and essential nutrients, brought to a boil, and sterilized in a large steam-heated vessel. Propagation media is transferred to cylindroconical propagation vessels (fermenters) and diluted to target density with sterile water.
2. The starter culture is added to the propagation medium.
3. The propagation is conducted at 24°C. Fermenters are aerated throughout propagation using sterile air.
4. Sugar consumption and pH are monitored throughout the growth cycle. When the culture has depleted all fermentable sugar, the temperature is lowered to 4°C and the cells are separated through natural processes of flocculation and settling. The yeast cell mass is collected as a concentrated liquid slurry.
5. Cell counts, viability measurements, and microbial QC are performed on all vessels.
6. Yeast is packaged in FDA-approved, sterilized HDPE containers and stored at 4°C until used to inoculate a beer fermentation. Containers are shipped overnight in an insulated box with ice packs.

To prevent contamination, the propagation medium is sterilized and all equipment is carefully cleaned and sanitized. Fermenters are cleaned in place with base and rinsed to neutral pH, then heat sanitized prior to filling with sterile propagation medium. Transfer lines are heat sanitized. Fermenters are sampled at many points throughout the process for quality control.

2.7. Product specifications

The specifications for liquid yeast are measured for every batch. All analytical methods used are validated for their intended use. If a lot of finished, food-grade liquid yeast slurry does not meet the quality control specifications, then the batch is rejected.

Viability is determined with an automated cell counter using AOPI fluorescence and with a hemocytometer using methylene blue staining. pH measurements are taken using a benchtop pH meter. Contaminating microorganisms are detected by swabbing to identify the presence of ATP and by plating samples on microbiological growth mediums, including non-differential medium (MYPG) and differential media (MYPG plus cycloheximide, MYPG plus copper, WLD, MRS). Lead levels are quantified using ICP-MS. Product specifications are listed in Table 1. Analytical results from five non-consecutive batches of BY-927 are listed in Table 2.

Parameter	Specification Range
Viable yeast cells	> 95%
Total bacterial count	0 per 1.00E+06 yeast cells
Total wild yeast count	0 per 1.00E+06 yeast cells

pH	< 4.6
Lead	< 10 µg/kg

Table 1. Liquid Yeast Specifications for BY-927.

Parameter	Batch A	Batch B	Batch C	Batch D	Batch E
Viable yeast cells	99.7%	99.7%	100%	99.6%	99.4%
Total bacterial count	0 per 1.00E+06 yeast cells	0 per 1.00E+06 yeast cells	0 per 1.00E+06 yeast cells	0 per 1.00E+06 yeast cells	0 per 1.00E+06 yeast cells
Total wild yeast count	0 per 1.00E+06 yeast cells	0 per 1.00E+06 yeast cells	0 per 1.00E+06 yeast cells	0 per 1.00E+06 yeast cells	0 per 1.00E+06 yeast cells
pH	4.2	3.99	3.97	4.1	3.94
Lead	< 10 µg/kg	< 10 µg/kg	< 10 µg/kg	< 10 µg/kg	< 10 µg/kg

Table 2. Analytical results from five non-consecutive batches of BY-927.

PART 3: DIETARY EXPOSURE

3.1. Intended use of BY-927

Berkeley Fermentation Science Inc. has proposed the use of BY-927 in brewing to reduce diacetyl production below sensory threshold levels (see Section 2.2). Diacetyl is a compound with a buttery, butterscotch-like flavor that has a detrimental impact on beer flavor and can be difficult and time-consuming to remove from beer. The yeast can be used as a drop-in replacement for other commercial yeasts in the beer manufacturing process.

3.2. Levels of use of the substance

BY-927 is used as any other commercial yeast would be used in the beer manufacturing process and is substitutional for any other *S. cerevisiae* strain currently used in beer fermentation. We recommend that BY-927 be inoculated at a pitch rate of $7.50\text{E}+05$ – $1.50\text{E}+06$ cells per mL per degree Plato of wort, as is standard industrial practice. Its use is not expected to increase dietary exposure to *S. cerevisiae*.

3.3. Dietary exposure

3.3.1. History of consumption

Saccharomyces cerevisiae has been used in food production for thousands of years, as discussed above in Section 2.3.1.

ALDC is widespread among acetic acid and lactic acid producing bacteria, including those used in food production (vinegar, yogurt, cheese, kombucha, beer, wine, cocoa processing), and may therefore be found in various food products. Purified ALDC enzyme is also commonly used as a processing aid in the brewing process to prevent diacetyl formation. TTB has formally approved an ALDC enzyme preparation produced by *Bacillus licheniformis* containing an ALDC gene from *Bacillus brevis* for use as a processing aid in the production of beer⁴¹. This enzyme is also the subject of a GRAS Notice (GRN No. 587) that received a "No Questions" letter from the FDA^{42,43}.

Diacetyl and its associated metabolites (acetoin and butane-2,3-diol) occur naturally in many foods and beverages common to the human diet^{44–46}. All three can be found in a variety of fermented products, including beer^{47,48}, wine⁴⁹, and cheese⁵⁰. Diacetyl and acetoin are also major contributors to the flavor and aroma profile of butter and are added to butter substitutes and various other foods to impart a characteristic butter flavor^{51,52}.

3.3.2. Estimated consumption

Exposure to the engineered yeast in finished beer will be extremely limited because yeast is removed from beer as part of the standard brewing process. Removing yeast from finished beer

is intrinsic to the brewing process: standard industrial brewer's yeast exhibit a strong propensity to sediment at the bottom of the fermentation vessel as fermentable sugars are exhausted. In a typical brewing process, the yeast is added to the wort and allowed to ferment at 64–74°C until fermentable sugars are fully consumed and standard gravity has stabilized. Once stabilized, the fermentation vessel is cooled to 0°C to promote sedimentation. After sedimentation is complete, the sedimented yeast is removed from the finished beer by racking the yeast from the bottom of the vessel. Using this method, only trace levels of yeast will remain in the finished beer. It is worth noting the common misconception that the haze in the hazy IPA beer style is due to unsedimented yeast; however, the haze is actually derived from high molecular weight complexes between proline-rich proteins and hop-derived polyphenols^{53–55}. Indeed, finished beer that contains residual yeast is considered a strong character flaw and a source of off-flavors. Additional methods of clarification include centrifugation and filtration^{56,57}, which remove all yeast that may be present. In light of this, it is extremely unlikely that beer made with BY-927 would contain a significant amount of yeast in solution.

When purified ALDC is added during the brewing process, it is not specifically removed from the final product. BY-927, however, produces the ALDC enzyme within the cell, and is removed from the final beer. Therefore, consumption of ALDC from beer produced with BY-927 will be negligible compared with beer made with purified ALDC, which is a standard brewing ingredient.

It is also worth noting that the only full-length yeast-derived proteins documented in finished beer are those that are cell-wall associated^{58–60}. In addition, glycolytic enzymes represent 30–60% of the total yeast proteome⁶¹ and ribosomal proteins represent another 30%^{62,63}, so proteins involved in other cellular functions are expressed at lower levels. Because the ALDC protein is not cell-wall associated, glycolytic, or ribosomal and is contained within the BY-927 cell cytoplasm, and because yeast is removed from beer before consumption, it is extremely unlikely that ALDC will be present in appreciable amounts in finished beer. Therefore, consumption of ALDC from beer produced with BY-927 will be negligible.

The modification to BY-927 results in lower diacetyl and lower diacetyl potential (the potential for excess α -acetolactate to form diacetyl after packaging). It does not result in the production of new compounds or increased concentrations of existing compounds. Similar to non-engineered strains, BY-927 reduces acetoin to butane-2,3-diol. The only difference is the time it takes to reach that endpoint, since non-engineered strains must first import diacetyl and reduce it to acetoin.

Beer produced with BY-927 will contain these compounds at concentrations that are similar to what is already present in consumed beer and similar to or lower than what is present in other foods and beverages, and therefore within levels safe for consumption. In order to quantify these compounds in beer made with BY-927, brewing trials were conducted at commercial scale and the beers were subject to analysis by liquid chromatography mass spectrometry (LC-MS) and gas chromatography mass spectrometry (GC-MS). The concentrations are listed in Table 3,

as well as the concentrations in commercial beers and other foods and beverages. See Section 6.7.3 for detailed information on the analytical methods.

Compound	Concentration in BY-927 beer	Concentration in commercial beers	Concentration in other foods and beverages
Diacetyl CAS No. 431-03-8	30 ± 7.7 µg/L	2–630 µg/L ^{64,65}	n.d.–2.5 mg/L (wine) ^{66,67} , 0.31–3.21 mg/L (butter) ⁶⁸ , up to 44 mg/L (baked goods) ⁶⁹
Acetoin CAS No. 513-86-0	1.25 ± 0.26 mg/L	0.2–3.9 mg/L ^{70–72} *	0.1–261 mg/L (wine) ^{49,73} , up to 350 mg/L (sherry) ⁷⁴
Butane-2,3-diol CAS No. 513-85-9	54.6 ± 0.02 mg/L	39–110 mg/L ⁷⁰	n.d.–2.3 mg/L (vinegar) ⁷⁵ , 0.16–3.2 mg/L (bananas) ⁷⁶

Table 3. Concentrations of target compounds in beer made with BY-927, commercial beers, and other foods and beverages. Beer made with BY-927 was subject to analysis by LC-MS and GC-MS. Concentrations in commercial beers and other foods and beverages were taken from the literature. n.d., not detected; * concentrations also measured in-house.

3.3.3. Toxicology and comparison data

Acetoin is a synthetic flavoring substance and adjuvant that is generally recognized as safe (GRAS) for its intended use by the FDA (21 CFR 182.60). Diacetyl is a substance permitted for direct addition to human food that is affirmed as GRAS (21 CFR 184.1278). Independent reviews conducted by both FEMA and the FAO/WHO determined that there is no safety concern regarding the use and consumption of these compounds based on current levels of intake. The judgment of the FEMA expert panel was that these substances are GRAS⁷⁷.

JECFA evaluated diacetyl and acetoin at its fifty-first meeting⁷⁸; the report states that although intake exceeds the threshold for Class II (540 µg/person/day), diacetyl and acetoin occur endogenously in humans and are metabolized to innocuous products. In addition, the intake of acetoin as a natural component of food is greater than that from its use as a flavoring agent. The report concluded that a NOAEL was not required for evaluation, and that both diacetyl and acetoin are of no safety concern based on current intake.

Butane-2,3-diol is not typically used as a flavoring agent and no repeated dose toxicity data is available. However, the European Chemicals Agency (ECHA) reported that no toxic effects were observed from a 2-year study in rats for the structural analogue butane-1,3-diol (NOAEL of 5000 mg/kg bw/d)⁷⁹. In addition, butane-2,3-diol is present as a natural component of beer and other

foods and beverages. JECFA also discussed butane-2,3-diol at its fifty-first meeting as a metabolic product of diacetyl and acetoin metabolism; the report states that all substances evaluated are metabolized to innocuous products and that these products, including butane-2,3-diol, are not of concern.

The concentrations of diacetyl, acetoin, and butane-2,3-diol in beer produced with BY-927 will be similar to or less than the levels in commercial beers, which are currently consumed (see Table 3 for a comparison). Therefore, we expect that any exposure to these compounds from beer made with BY-927 is substitutional. Based on this, we can conclude that exposure to these compounds is unlikely to cause a safety concern.

PART 4: SELF-LIMITING LEVELS OF USE

There are no proposed restrictions on the use of BY-927.

Use of BY-927 requires no alterations to the standard pitch rate - the amount of yeast added to wort to start a fermentation. Regardless of pitch rate, the quantity of yeast cells present at the end of the fermentation should not be significantly different. In addition, yeast is removed from finished beer prior to packaging, leaving only trace levels in the finished product.

PART 5: EXPERIENCE BASED ON COMMON USE IN FOOD BEFORE 1958

This part does not apply.

PART 6: NARRATIVE

6.1. Overview of safety assessment

In this section, we present evidence that the modified *S. cerevisiae* strain, BY-927, is safe for use in industrial brewing. BY-927 produces cleaner tasting beer by preventing the formation of the common off-flavor diacetyl. Diacetyl concentrations are kept below sensory threshold throughout fermentation and after packaging. We detail the safety of the host and donor strains, the ALDC enzyme, the gene regulatory regions, and the small molecule composition of beer made with BY-927, using both targeted and untargeted methods.

6.2. Safety assessment of the host strain

The host is an industrial brewing strain of *S. cerevisiae* that has been used extensively in commercial beer production for decades. *S. cerevisiae* has an extensive history of safe use; it has been used for more than 7,000 years by humans in fermentation processes, such as bread leavening and beer and wine production. Fermentation products of this yeast are not associated with adverse effects in humans (when consumed in recommended quantities for alcohol intake).

S. cerevisiae is considered GRAS through its use in the brewing, baking, and winemaking industries. In the 27th report of The Scientific Committee for Human Food of the European Community, the authors state that *Saccharomyces cerevisiae* has a safe history of use in food and does not produce toxins.

6.3. Safety assessment of the donor strains

Donor strain *Acetobacter aceti* is a benign, gram-negative bacterium that has an extensive history of safe use. Acetic acid bacteria, including *Acetobacter aceti*, have been used in vinegar production for thousands of years. The first recorded evidence of vinegar production dates to around 3000 B.C. from the Ancient Babylonians although spontaneous fermentations (including the souring of wine) were likely happening far earlier. Members of the *Acetobacter* genus, including *A. aceti*, are still used in commercial vinegar production today. *A. aceti* is on the FDA's list of GRAS microorganisms; the EPA has included it as exempt from review under certain use conditions in the Toxic Substances Control Act given its long history of safe use and its classification as a low risk organism (CFR 725.420). There are no reports suggesting that *A. aceti* is pathogenic to humans or animals, despite it often coming into contact with humans. It does not produce toxins, enzymes, or virulence factors associated with pathogenicity or allergenicity.

The safety assessment of donor strain *S. cerevisiae* has been previously established.

6.4. Safety assessment of the ALDC enzyme

Source: *Acetobacter aceti*

Enzyme name: α -Acetolactate decarboxylase

Reaction: α -acetolactate $\rightarrow$ acetoin + CO₂

ALDC catalyzes the conversion of α -acetolactate to acetoin, thereby circumventing diacetyl formation and removing excess α -acetolactate which has the potential to form diacetyl after packaging. The ALDC from *A. aceti* is well characterized and has previously been shown to catalyze this reaction; it does not encode some other activity³⁵.

This enzyme is widespread among acetic acid and lactic acid bacteria, several species of which are used in food production. Purified ALDC enzyme is also commonly used as a processing aid in the production of beer, and several enzyme preparations derived from food-grade bacteria are commercially available for this purpose⁸⁰. One of these, derived from *Bacillus brevis*, is the subject of a GRAS Notice (GRN No. 587) that received a "No Questions" letter from the FDA.

6.4.1. Allergenic/Toxicogenic potential of the ALDC enzyme

It is extremely unlikely that intracellular ALDC could cause a food allergy. This is based on the following considerations:

1. The vast majority of enzymes present in the natural world are not allergens, and a wide variety of enzyme classes and structures are naturally present in plant and animal based foods.
2. The concentration of ALDC in finished beer is anticipated to be nominal (see Section 3.3.2). The total amount of ALDC protein is a small fraction of the total yeast protein and because the ALDC is expressed inside the cell, and because the yeast is removed from beer before consumption, only trace levels of ALDC protein or peptide fragments will be found in finished beer.
3. This enzyme is derived from a microorganism that has been used in food production for centuries. In addition, the ALDC gene is widespread among acetic acid and lactic acid bacteria¹⁸, so ALDC may be found in food products such as vinegar, yogurt, and cheese. Therefore, the enzyme already has a history of safe consumption.
4. Purified ALDC is used in commercial beer production. Therefore, the enzyme already has a history of safe consumption and is already present in the brewing process.

In order to further evaluate the possibility that the ALDC enzyme will induce a reaction in an already sensitized individual, amino acid sequence similarity to known allergens was assessed. Following the guidelines developed by FAO/WHO, 2001^{81, 82} and modified by Codex Alimentarius Commission, 2009⁸³, the ALDC amino acid sequence was compared to allergens

from the Food Allergy Research and Resource Program (FARRP) allergen protein database (<http://allergenonline.org>) as follows:

We conducted an alignment of the expressed protein to each of the allergens in the FARRP. E-values < 0.05 were considered significant. No significant similarity was found between the ALDC amino acid sequence and any of the allergens from the FARRP database. This was repeated using a sliding window of 80 amino acid segments of the ALDC amino acid sequence; no matches with more than 35% identity were found. In addition, a search for 100% identity over 8 contiguous amino acids was completed; no matches were found.

We next set out to determine whether the expressed enzyme is similar to known toxins, venoms, or virulence factors. We extracted all peptide sequences in the UniProt and Virulence Factor databases described in Table 5, then queried these sequences with the expressed ALDC protein using the BLASTP algorithm and default parameter settings⁸⁴. In total, this queried more than 142,000 sequences. E-values < 0.05 were considered significant. All hits obtained during this search had E-values > 0.05, indicating that the alignments were due to short random sequence identities and that ALDC is not homologous to any of the annotated toxins, venoms, or virulence factors in these databases.

In addition to the assessments based on sequence alignment, we also analyzed the ALDC protein using ToxinPred2^{85,86}, a web-based prediction tool for protein toxicity, set to default parameter settings. This tool uses a hybrid approach combining a similarity-based approach using BLAST, a motif-based approach using MERCI, and a machine learning approach to predict protein toxicity. Based on the overall score, which is computed by integrating the three different methods, the protein is categorized as toxic or nontoxic. The prediction was that ALDC is nontoxic.

On the basis of the available evidence, we concluded that the ALDC protein does not pose any food allergenic or toxigenic concerns.

Database Name	Database description	Number of entries in database
Uniprot/Swiss-Prot toxin/venom database	Manually curated list of proteins that are produced by any organism and identified as either venoms or toxins	8,743
Uniprot toxin/venom database	Uniprot complete toxin/venom database. List of proteins that are produced by any organism and identified as either venoms or toxins. This includes sequences that are manually curated as well as those that are unreviewed and automatically curated.	101,389
Virulence Factor database	VFDB core dataset; includes genes	4,252

(VFDB) core protein set	associated with experimentally verified virulence factors	
Virulence Factor database (VFDB) full protein set	VFDB full dataset; covers all genes related to known and predicted virulence factors	27,982

Table 4. Toxin, venom, and virulence factor databases.

6.5. Safety assessment of the promoter and terminator sequences

The promoter and terminator sequences were isolated from *S. cerevisiae*. The safety assessment of *S. cerevisiae* has already been established and since the regulatory sequences do not code for proteins, the source of this material does not raise a safety concern.

6.6. Presence of unintended gene products

As an additional measure, we tested whether the insert contained open reading frames (ORFs) beyond the one that was intentionally introduced. We searched for ORFs of more than 100 codons using the NCBI ORF Finder (<https://www.ncbi.nlm.nih.gov/orffinder/>). Analysis using the NCBI ORF Finder shows that the integrated region contains one putative ORF. However, it is contained within the promoter and therefore corresponds to a sequence already present in *S. cerevisiae*. In the unlikely event that this ORF is translated into a protein, it would be found in both BY-927 and the parent strain. As the parent strain has never been shown to release unwanted products corresponding to this sequence, neither should BY-927.

6.7. Safety assessment of BY-927

6.7.1. Safety of BY-927

BY-927 is classified as *S. cerevisiae*, a species that is known to be non-pathogenic and safe for use in industrial food and beverage processes. BY-927 is non-pathogenic and non-toxicogenic. No genes encoding for virulence factors, protein toxins, or enzymes involved in the synthesis of toxins, allergens, or other undesirable substances are expected based on our knowledge of the strain, the gene sequences, and the promoters and terminators.

6.7.2. Changes in brewing procedures or beer composition as a consequence of BY-927 fermentation

The use of BY-927 does not require changes to brewing procedures. BY-927 functions as a drop-in replacement for beer created with any industrial brewing strain, in that a fermentation performed with BY-927 will be substantially equivalent to a standard fermentation.

The only difference between BY-927 and a non-engineered strain is the diacetyl profile during fermentation. BY-927 consistently keeps diacetyl levels low (and well below the flavor threshold) throughout fermentation, while non-engineered strains produce a spike in diacetyl levels at the

start of fermentation. By the end of primary fermentation, though, diacetyl levels are similar. If purified ALDC is added at the start, the diacetyl profile during primary fermentation will be similar to that of BY-927. However, a significant difference is noticeable after dry hopping: while BY-927 maintains diacetyl levels below the flavor threshold, non-engineered strains produce a secondary increase in diacetyl, prolonging the brewing process as brewers wait for levels to decrease.

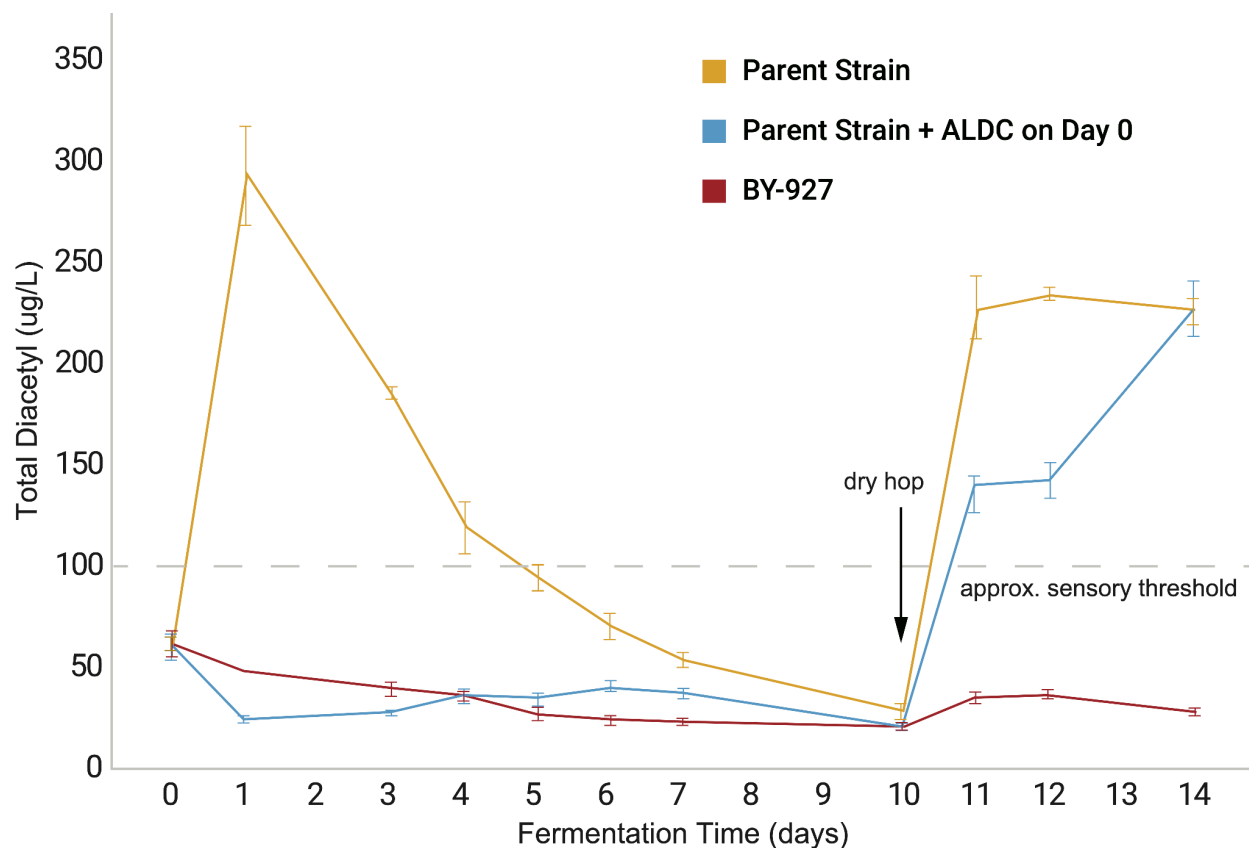


Figure 4. Performance of BY-927 compared to its non-engineered parent and purified ALDC enzyme. The parent strain shows a typical diacetyl profile, with diacetyl levels increasing at both the start of fermentation and after dry hopping. Adding ALDC at the start of fermentation reduces diacetyl during primary fermentation, but the enzyme is no longer effective after 10 days and diacetyl spikes after dry hopping. BY-927 consistently keeps diacetyl levels well below the sensory threshold, even after dry hopping.

6.7.3. Identification and quantification of diacetyl, acetoin, and butane-2,3-diol in beer fermented with BY-927

Beer made with BY-927 was subject to diacetyl analysis by liquid chromatography mass spectrometry (GC-MS) using the following method: yeast-free beer samples were combined with 10X phosphate buffered saline at pH 7.4 and 50 mg/L 3,4-diaminopyridine, which acted as a derivatizing agent. The samples were incubated at 60°C for 1 hour, then extracted with ethyl acetate. The resulting ethyl acetate was dried under vacuum, then subsequently dissolved in

HPLC-grade water and transferred into glass sample vials for analysis by LC-MS. Samples were chromatographed at 30°C in a reversed-phase C18 column (2.1 x 5.0 mm x 1.8 µm) installed in an Agilent 1290 UHPLC and coupled to an Agilent 6130 single-quadrupole mass spectrometer. Metabolites were separated using a linear gradient of 3-10% acetonitrile supplemented with 0.1% (v/v) formic acid over 6 minutes followed by a wash at 95% and reequilibration at 3%. Diacetyl was detected by mass spectrometry with selected ion monitoring (SIM) at specific *m/z* values. We confirmed the chemical identity of diacetyl based on the injection of an authentic standard, which matched perfectly in retention time and in mass spectrum. Quantification was accomplished by comparing peak areas to standard curves generated by spiking known concentrations of isotopically labeled diacetyl (diacetyl-d6) into a beer matrix, then following the same derivatization and extraction protocol (Figure 5).

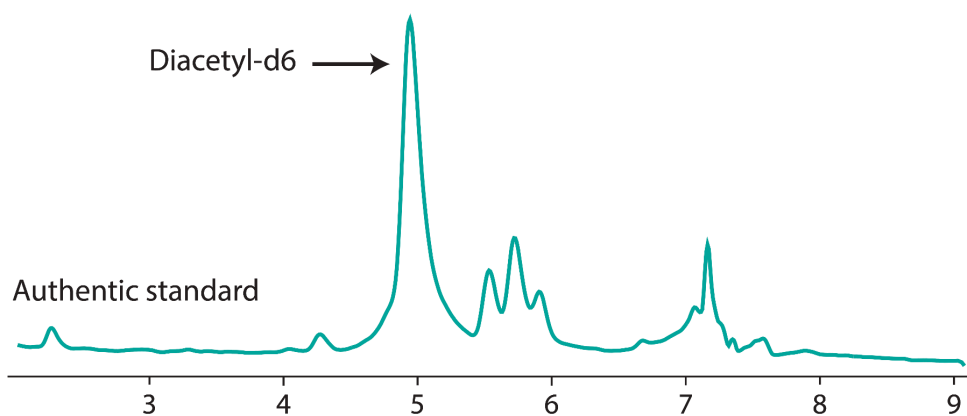
Beer made with BY-927 was subject to acetoin and butane-2,3-diol analysis by gas chromatography mass spectrometry (GC-MS) using the following method: yeast-free beer samples were pH adjusted, dosed with 2-octanol as an analytical internal standard, and extracted with ethyl acetate. The resulting ethyl acetate was treated with sodium sulfate to remove excess water, then injected onto an Agilent 6890 gas chromatograph fitted with a high-polarity, ultra-inert column (Agilent G3903-63008) and coupled to an Agilent 5973N single-quadrupole mass analyzer. Each sample was resolved over a temperature ramp from 40°C to 250°C over 26.6 min, with a total per-sample run time of 33.63 min including low and high temperature holding time. Compounds were detected by mass spectrometry with selected ion monitoring (SIM) at specific *m/z* values. We confirmed the chemical identities of acetoin and butane-2,3-diol based on injections of authentic standards, which matched perfectly in retention time and in mass spectrum. Quantification was accomplished by comparing peak areas to standard curves generated by spiking known concentrations of target compounds into a beer matrix, then following the same extraction protocol (Figure 6).

Compound	Concentration
Diacetyl	30 ± 7.7 µg/L
Acetoin	1.25 ± 0.26 mg/L
Butane-2,3-diol	54.6 ± 0.02 mg/L

Table 5. Quantification of target compounds.

Selected ion monitoring chromatograms, LC-MS

a



b

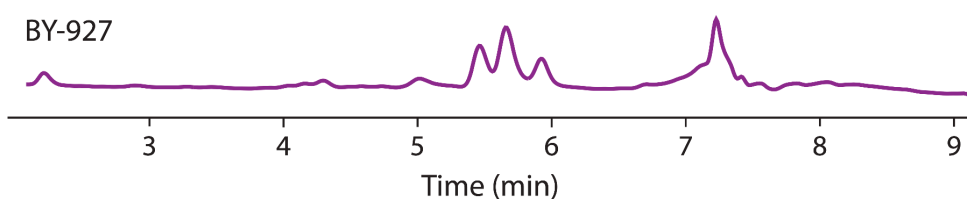


Figure 5. Quantification of diacetyl using liquid chromatography mass spectrometry. SIM chromatograms from a diacetyl-d6 authentic standard (teal) and a BY-927 fermentation sample (purple) are shown. Diacetyl d-6 was spiked into a standard beer matrix, fermented with a non-engineered strain. Samples were prepared using the method described above. The y-axis represents signal intensity in arbitrary units. Additional standard curve traces are not shown.

Selected ion monitoring chromatograms, GC-MS

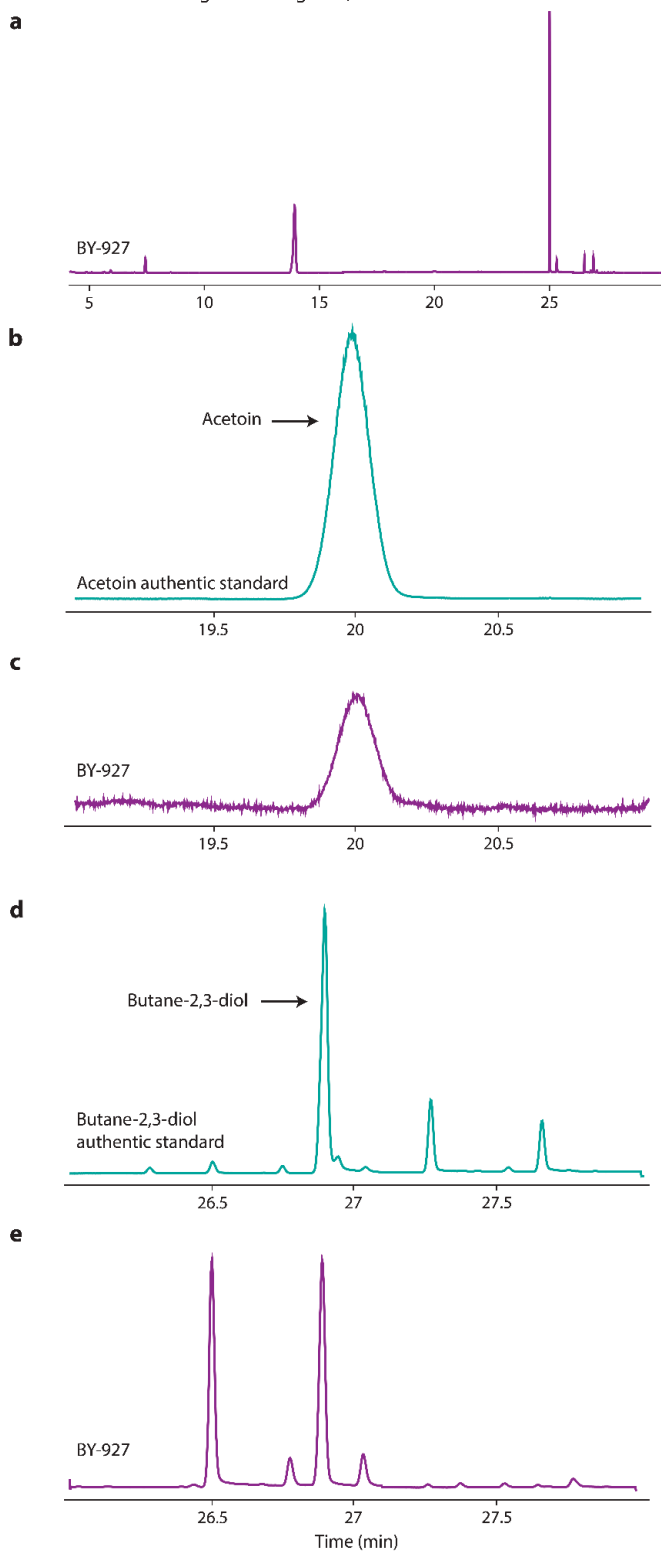


Figure 6. Quantification of acetoin and butane-2,3-diol using gas chromatography mass spectrometry. SIM chromatograms from BY-927 fermentation samples (purple) and acetoin and

butane-2,3-diol authentic standards (teal) are shown as follows: (a) The entire BY-927 chromatogram; (b) minutes 19-21 of the authentic standards chromatogram showing acetoin; (c) minutes 19-21 of the BY-927 chromatogram; (d) minutes 26-28 of the ester authentic standards chromatogram showing butane-2,3-diol; (e) minutes 26-28 of the BY-927 chromatogram. Authentic standards were spiked into a standard beer matrix, fermented with a non-engineered strain. Samples were prepared using the method described above. The y-axis represents signal intensity in arbitrary units. Additional standard curve traces are not shown.

6.7.4. Untargeted metabolomic profiling of BY-927

The modifications resulted, as expected, in reduced diacetyl. However, unexpected effects are also possible. In order to evaluate whether the genetic modification introduced unexpected small molecule production that might pose a risk for consumption, un-targeted metabolomics was performed on beer produced with BY-927, its non-engineered parent strain, and three other commonly-used industrial brewing strains. We analyzed these samples by gas chromatography mass spectrometry (GC-MS) to measure volatile molecules and by electrospray ionization liquid chromatography mass spectrometry (ESI-LC-MS) in positive ion detection mode to measure non-volatile molecules.

To perform the GC-MS analysis, quadruplicate cultures of the strains listed above were inoculated into 80 mL of malt extract. Anaerobic fermentation then proceeded over 5 days at 20°C. The additional non-engineered yeast strains were included in this study to help benchmark the variation in small molecule concentrations without genetic engineering. Upon completion of fermentation, the yeast-free beer-like samples were pH adjusted, dosed with 2-octanol and 12-[(cyclohexylamino)carbonyl]amino]-dodecanoic acid as analytical internal standards, then extracted with ethyl acetate. The resulting ethyl acetate was treated with sodium sulfate to remove excess water, then injected onto an Agilent 6890 GC fitted with a high-polarity, ultra-inert column (Agilent G3903-63008) and coupled to an Agilent 5973N single-quadrupole mass analyzer. Each sample was resolved over a temperature ramp from 40°C to 250°C over 26.25 min, with a total per-sample run time of 35.25 min including low and high temperature holding time. The mass detector collected signals ranging from 30 to 300 *m/z*. Raw signal was normalized to the intensity of the 2-octanol internal standard before further processing. Figure 7 shows the total ion chromatograms of a representative sample from each of BY-927 and its non-engineered parent, showing that the two ferments are highly similar.

For more comprehensive data analysis the raw GC-MS data files were uploaded onto the XCMSOnline server (https://xcmsonline.scripps.edu/landing_page.php?pgcontent=mainPage)⁸⁷, which automates retention time alignment, peak alignment, intensity normalization, and then identifies statistically significant, intensity-differentiated mass features. The preset “GC-MS / single quad” parameter was selected for data processing, which yielded 127 mass features among the five yeast strains upon multigroup analysis. We then searched for differentially increased molecules in BY-927 compared to its parent in paired analysis while applying three selection filters: (1) q-value less than 0.05, (2) peak intensity fold change less than 3, and (3) fold-change over the non-engineered parent greater than two times the maximum intensity difference among any pair of non-engineered yeast strains. This third filter is designed to

eliminate features that may seem differentially increased in BY-927 but are not biologically significant when considering existing variation among non-engineered strains. Note, however, that for this dataset, the third filter was not necessary because there were no features remaining after filters 1 and 2 were applied. We did not detect any statistically and biologically significant mass features increased in BY-927 compared to its parent. We concluded that GC-MS analysis did not find molecules that warranted further safety assessment.

To perform the ESI-LC-MS analysis, fermented samples created in the same method as above were processed through C₁₈ solid phase extraction columns (Waters Sep-Pak WAT020515). Columns were equilibrated with deionized water, then loaded with 5 mL of sample. After the samples were pulled through by vacuum, the columns were rinsed with 5 mL deionized water and eluted with 3 mL methanol. The eluate was dried overnight under vacuum to leave behind a dark yellow film, then subsequently dissolved in 5% acetonitrile. 5 μ L of this solution was injected onto the Agilent 1290 Infinity II LC equipped with a C₁₈ column (Agilent 959757-902) coupled to a C₁₈ guard column (Agilent 821725-901). The mobile phase was dosed with 0.1% formic acid and ramped from 5% acetonitrile in water to 30% acetonitrile over 30 min, with a total per-sample run time of 45 min including holding time. The Agilent 6130 single-quadrupole mass detector collected signals ranging from 100 to 1,000 *m/z*. All sample vials were held at 4°C in the autosampler throughout the duration of the queue. Figure 8 shows the total ion chromatograms of a representative sample from each of BY-927 and its parent, showing that the two ferments are highly similar. This result is expected because features derived from wort which are not altered by yeast fermentation contribute a larger percentage of the signal in LC-MS data.

For data analysis, the raw LC-MS data was uploaded onto the XCMS online server and underwent the same pre-processing steps as listed above for the GC-MS data, using the preset “HPLC / single quad” parameter. Among the five yeast strains, there were 837 mass features detected. Here a mass feature is defined as a unique [*m/z*, retention time] pair. After applying the same statistical and biological thresholds as above, we did not detect significantly increased features in BY-927 compared to its parent. We concluded that the LC-MS analysis did not find molecules that warranted further safety assessment.

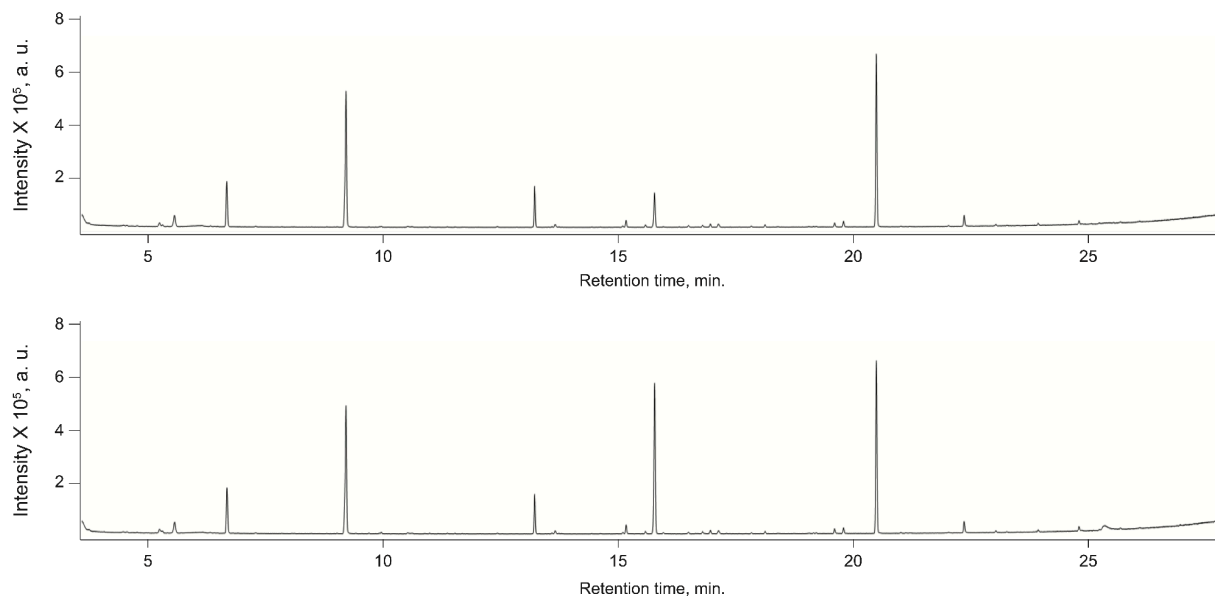


Figure 7. Representative GC-MS total ion chromatograms of BY-927 (top) and its non-engineered parent (bottom).

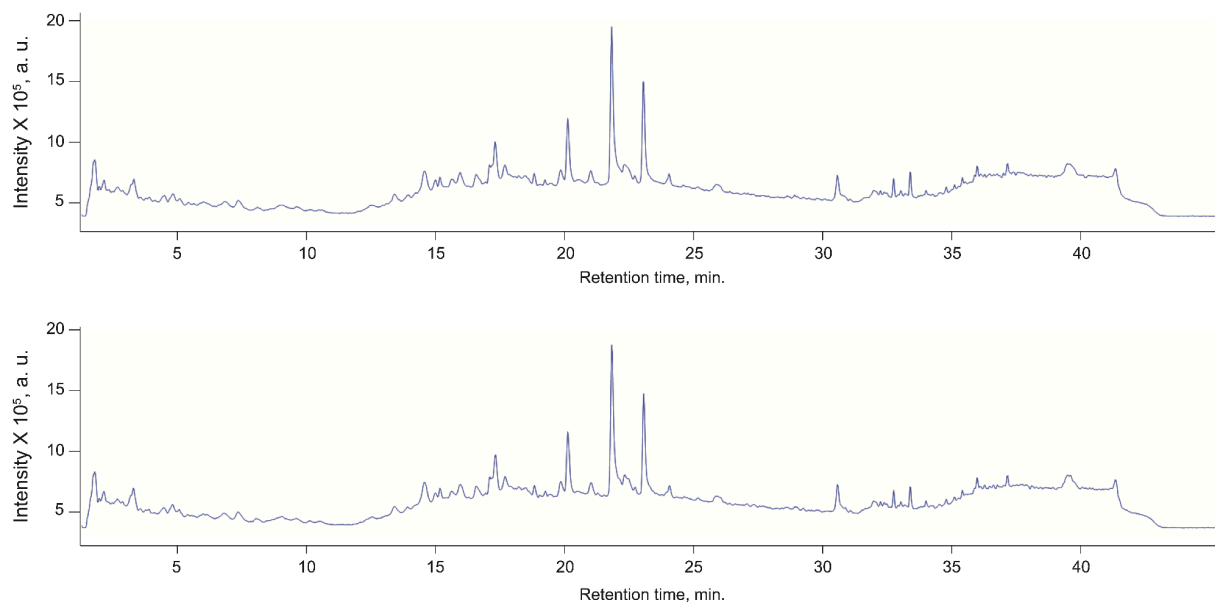


Figure 8. Representative LC-MS total ion chromatograms of BY-927 (top) and its non-engineered parent (bottom).

6.8. Precedence for safety of engineered *S. cerevisiae* strains

The FDA has had no questions on GRAS notifications for a number of *S. cerevisiae* strains where heterologous genes were inserted or endogenous genes were edited so as to eliminate undesirable compounds or to produce compounds known to be safe. Examples include

Lesaffre's malolactic strain ML01 (GRN No. 120), Phyterra's urea degrading strain ECMo01 (GRN No. 175), Phyterra's low hydrogen sulfide strain P1Y0 (GRN No. 350), Mascoma's lactic acid-producing strain (GRN No. 841), Berkeley Fermentation Science's monoterpene-producing strain yBBS002 (GRN No. 798) and volatile thiol-releasing strain BY-989 (GRN No. 1094), and Omega's POF- strain OYR-185 (GRN No. 1062) and volatile thiol-releasing strain OYR-243 (GRN No. 1096).

Recently, Omega's ALDC-expressing strain OYR-481 received a "No Questions" letter from the FDA (GRN No. 1190). This strain is also a modified *S. cerevisiae* brewing strain used in brewing to prevent the formation of diacetyl. Like BY-927, OYR-481 expresses an α -acetolactate decarboxylase and catalyzes the conversion of α -acetolactate to acetoin.

6.9. Conclusions for GRAS determination

Berkeley Fermentation Science Inc. has made the following conclusions regarding the use of yeast strain BY-927 as a processing aid to manufacture beer:

- *S. cerevisiae* has a long history of safe use in food and beverage production. Across a variety of use cases, *S. cerevisiae* and *S. cerevisiae*-derived products have been approved as food additives, affirmed as GRAS substances, and the subject of previous GRAS Notices.
- BY-927 is derived from an industrial brewing strain of *S. cerevisiae* that is used extensively in commercial beer production. The strains are substantially equivalent except for the introduction of a single gene, which confers the ability to convert α -acetolactate directly into acetoin, circumventing diacetyl formation.
- BY-927 is genetically stable. Industrial brewing yeast are generally tetraploid, and the modifications have occurred at the target locus across four homologous chromosomes. DNA diagnostic methods confirmed the presence of the insert and the genetic similarity between BY-927 and its parent.
- The engineering does not result in the production of new compounds or increased concentrations of existing compounds. The concentrations of diacetyl, acetoin, and butane-2,3-diol in beer produced with BY-927 will be similar to or less than the levels in commercial beers. Therefore, we expect that any exposure to these compounds from beer made with BY-927 is substitutional.
- Diacetyl and acetoin are flavoring agents that are generally recognized as safe for their intended use and permitted for direct addition to food by the FDA. Independent reviews conducted by both FEMA and the FAO/WHO determined that there is no safety concern regarding the use and consumption of these compounds based on current levels of intake. Diacetyl and acetoin are metabolized to butane-2,3-diol, which is innocuous and not of concern.

- The ALDC enzyme present in BY-927 is derived from a bacterium that has been used in food production for centuries. It does not code for either toxic or allergenic proteins, nor is its enzymatic activity implicated in the formation of unanticipated compounds.
- ALDC is widespread among acetic acid and lactic acid bacteria, several species of which are used in food production. ALDC may be found in food products such as vinegar, yogurt, and cheese. Therefore, the enzyme already has a history of safe consumption.
- Purified ALDC enzyme is commonly used as a processing aid in commercial beer production to convert secreted α -acetolactate into acetoin. Therefore, the enzyme is already present in the brewing process. Several enzyme preparations derived from food-grade bacteria are commercially available for this purpose, including one derived from *Bacillus brevis* that is the subject of a GRAS Notice (GRN No. 587) that received a "No Questions" letter from the FDA.
- BY-927 is removed from beer as part of the standard brewing process; therefore only trace levels of BY-927 remain in the finished beer and dietary exposure to BY-927 is negligible.
- Un-targeted comparative metabolomics was performed on beer produced with BY-927 and its parent strain. GC-MS and LC-MS analysis did not detect any statistically and biologically significant mass features in BY-927 that warranted further safety assessment.

Berkeley Fermentation Science Inc. has determined, through the scientific analysis presented in this notice, that yeast strain BY-927 is "Generally Regarded as Safe" (GRAS) for use in brewing and is, therefore, exempt from premarket approval requirements. Additionally, Berkeley Fermentation Science Inc. has reviewed the available data and is not aware of any information that would be inconsistent with the GRAS conclusion.

PART 7: SUPPORTING DATA AND INFORMATION

7.1. References

The details/parameters of our review of available scientific literature/data are as follows:

- Key search terms: diacetyl, α -acetolactate, acetoin, butane-2,3-diol, valine, vicinal diketone, beer, beverages, α -acetolactate decarboxylase
- Document types: reviews, primary research, toxicology studies, government documents
- Our review of available scientific literature/data is complete and balanced as it covers six decades of research up to today, and provides a complete understanding of the function of the expressed enzyme and the presence, origin, and concentration of diacetyl, acetoin, and butane-2,3-diol in beer and other fermented beverages.

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DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved:OMB No. 0910-0342; Expiration Date: 08/31/2025 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER	DATE OF RECEIPT
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration,5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____ ☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3. Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): 2025-01-06	
4. For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person Charles Denby		Position or Title CEO	
	Organization (<i>if applicable</i>) Berkeley Fermentation Science Inc.			
	Mailing Address (<i>number and street</i>) 15555 E 14th St Ste 525			
City San Leandro		State or Province California	Zip Code/Postal Code 94578	Country United States of America
Telephone Number 510-900-7231		Fax Number	E-Mail Address charles@berkeleyyeast.com	
1b. Agent or Attorney (<i>if applicable</i>)	Name of Contact Person		Position or Title	
	Organization (<i>if applicable</i>)			
	Mailing Address (<i>number and street</i>)			
City		State or Province	Zip Code/Postal Code	Country
Telephone Number		Fax Number	E-Mail Address	

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Saccharomyces cerevisiae strain BY-927

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway ☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)* ☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN _____

☐ b) GRAS Affirmation Petition No. GRP _____

☐ c) Food Additive Petition No. FAP _____

☐ d) Food Master File No. FMF _____

☐ e) Other or Additional *(describe or enter information as above)* _____

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))* ☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? *(see 21 CFR 170.225(c)(8))*

☐ Yes *(Proceed to Item 8)*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

Saccharomyces cerevisiae strain BY-927 is intended for use in brewing to improve beer quality by preventing the formation of the off-flavor compound diacetyl. Diacetyl is a compound with a buttery, butterscotch-like flavor that has a detrimental impact on beer flavor and can be difficult and time-consuming to remove from beer. BY-927 contains a gene from Acetobacter aceti, a food microorganism used in commercial vinegar production, that encodes an α -acetolactate decarboxylase enzyme (ALDC). ALDC converts the diacetyl precursor, α -acetolactate, directly into acetoin. We recommend that BY-927 be inoculated at a pitch rate of $7.50E+05$ – $1.50E+06$ cells per mL per degree Plato of wort. Its use is not expected to increase dietary exposure to *S. cerevisiae*.

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes ☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☐ Yes ☒ No

Did you include this other information in the list of attachments?

☐ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Berkeley Fermentation Science Inc.

(name of notifier)

has concluded that the intended use(s) of BY-927

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Berkeley Fermentation Science Inc. agrees to make the data and information that are the basis for the
(name of notifier) conclusion of GRAS status available to FDA if FDA asks to see them;
agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

15555 E 14th St Ste 525, San Leandro CA 94578

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Charles Denby

Digitally signed by Charles Denby
Date: 2025.06.09 12:53:42 -07'00'

Printed Name and Title

Charles Denby, CEO

Date (mm/dd/yyyy)

06/09/2025

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	CoverLetter_BY-927_2025-06-09.pdf	Administrative
	GRASNotice_BY-927_2025-06-09.pdf	GRAS Notice

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRASStaff@fda.hhs.gov. (Please do NOT return the form to this address.). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.